

HAT-P-54b

R-1417

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_200211 · created 2026-08-01T05:56:07+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458890.8064 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.88 +/- 2.26 [%]
Orbital Inclination (inc)	83.64 +/- 2.3
Airmass coefficient 1 (a1)	1.02 +/- 0.89
Airmass coefficient 2 (a2)	0.24 +/- 0.89
Scatter in the residuals of the lightcurve fit is	135.25 %
Best Comparison Star	#3 - [158, 422]
Optimal Aperture	1.73
Optimal Annulus	13.97
Transit Duration (day)	0.044 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.057 vs 0.1572 (deviation 0.68σ)
Duration fit vs published	0.044 d vs 0.075 d (deviation 1.21σ)
Mid-time O–C	2.5 min
Depth SNR	3.4
Residual scatter	135.25 %

Light curve

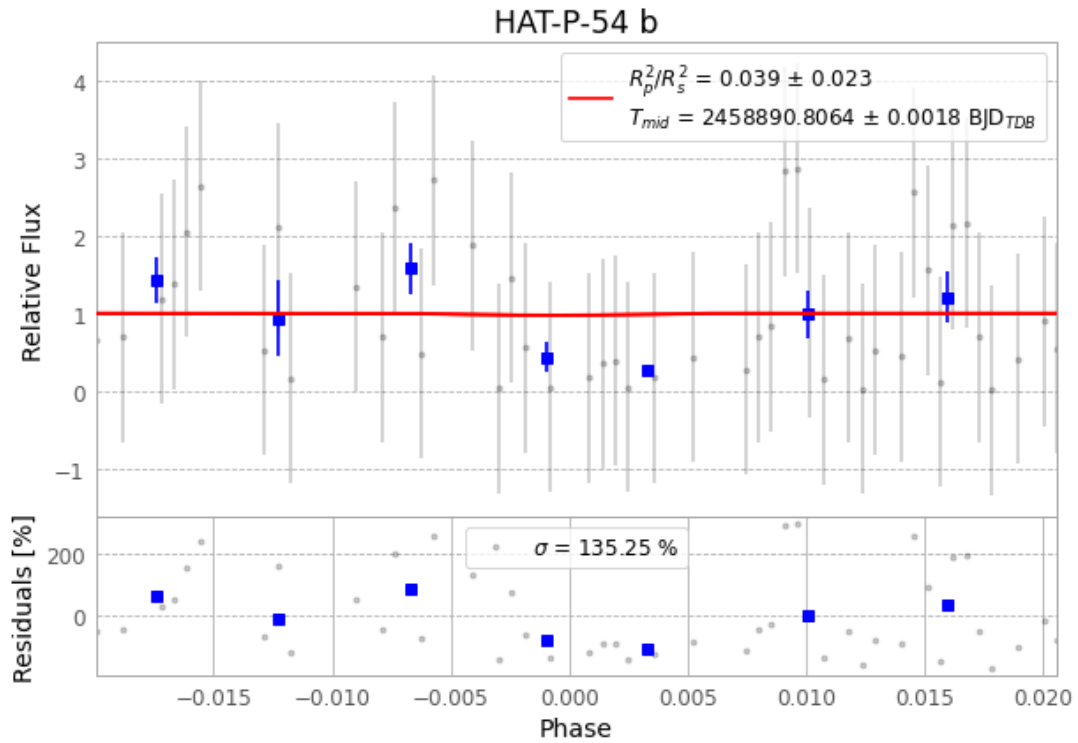


Figure 1 — FinalLightCurve_HAT-P-54 b_2020-02-10.png (SHA-256 8ddb66b98cc7cd3b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9cd9daa2a733ee21...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), HATP-54200211052112.FITS → HATP-54200211090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-200211.log (3fc990c3d3c5...), FinalLightCurve_HAT-P-54 b_2020-02-10.png (8ddb66b98cc7...), FinalLightCurve_HAT-P-54 b_2020-02-10.csv (efbf9ffa5222...)

Compute resources

Wall time 285.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 05:56Z.